

# BelkaCTF #7 — Compromised Air Traffic Control Workstation

Case INC-2026-0624-0001 · Examiner: niel · 2026-06-24 21:36 UTC

4

FINDINGS

1

EVIDENCE

ITEMS

10

IOCS



## CLASSIFICATION

**CONFIDENTIAL — AVIATION SAFETY IMPLICATIONS**

## TARGET SYSTEM

**TSO-ATC-CT412 (Windows 10 build 19041  
(19041.1.amd64fre.vb\_release.1912))**

## COMPROMISE DETECTED

**YES — 3 CRITICAL severity findings, active C2  
established**

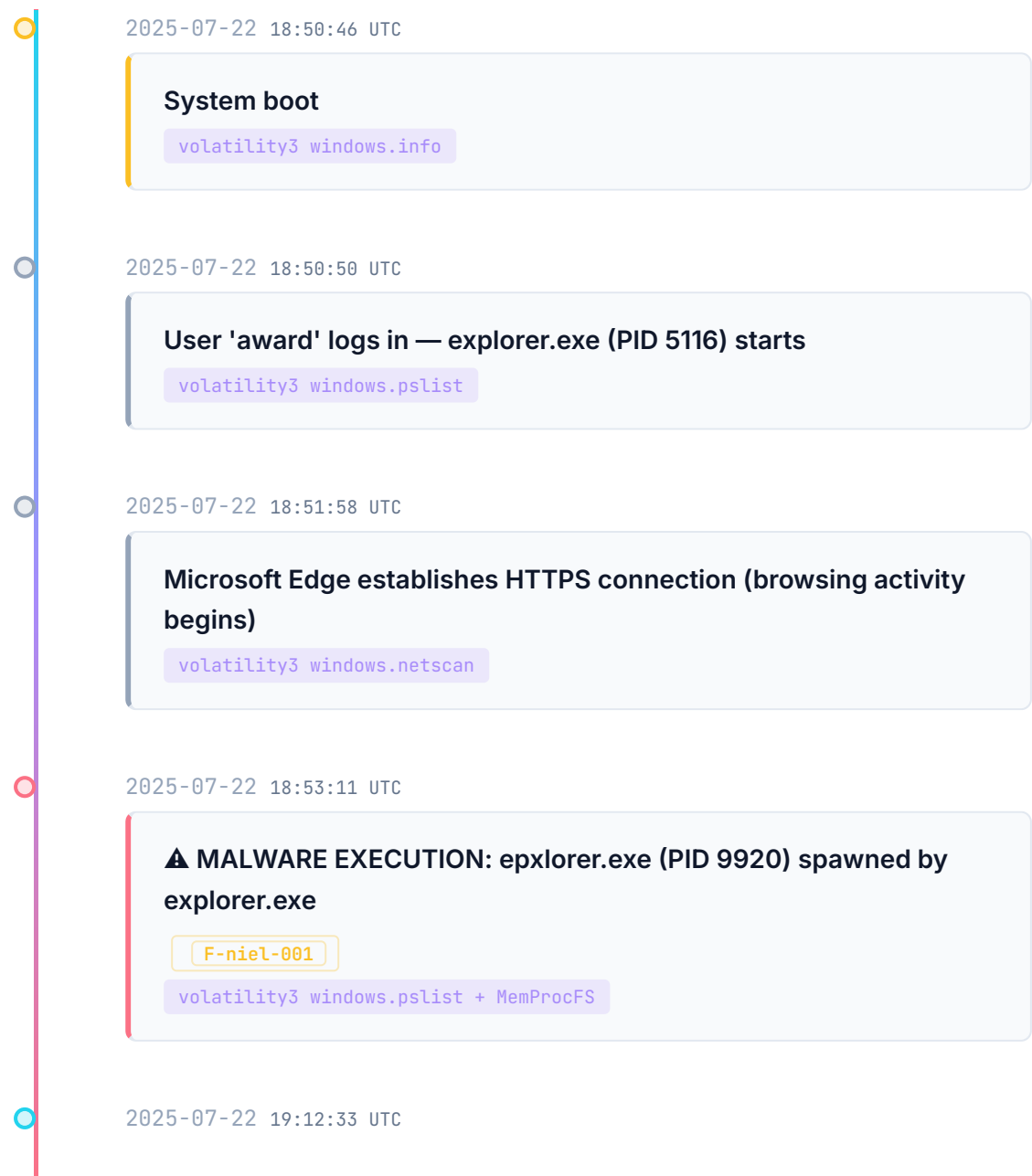
## INVESTIGATION WINDOW

**2025-07-22 18:50 → 19:18 UTC (28 minutes)**

On 2025-07-22, an air traffic control workstation (TSO-ATC-CT412, user 'award') was compromised within 3 minutes of user login. A phishing email from ktso.sec@mailforce.net delivered a trojan (epxplorer.exe) masquerading as the Windows

shell process explorer.exe. The malware established command-and-control to an attacker-operated Telegram Bot API behind Cloudflare infrastructure within 19 minutes of execution. At the time of compromise, ATC\_RADAR.exe — an aviation radar processing application — was actively running, meaning unauthorized code coexisted with live air traffic operations. This constitutes a potential aviation safety incident with possible regulatory reporting obligations under ICAO Annex 17. Immediate network isolation of the affected workstation and escalation to the aviation safety officer are required.

Chronological reconstruction of events from memory forensics artifacts. Color-coded by event type. Hover cards for source details.



● C2 ESTABLISHED: epxlorer.exe → 104.21.1.247:443 (Telegram Bot API)

F-niel-002

volatility3 windows.netscan



2025-07-22 19:15:05 UTC

ATC\_RADAR.exe active — malware coexisting with aviation operations software

F-niel-004

volatility3 windows.pslist + MemProcFS



2025-07-22 19:18:53 UTC

Memory dump captured — evidence acquisition point

volatility3 windows.info

- Malware / Compromise
- Network / C2
- Process Activity
- File System
- System Event
- User Activity



| ID         | FINDING                                                                                                                                                                                                                      | SEVERITY /<br>CONFIDENCE | MITRE<br>ATT&CK        | TOOL                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------|-------------------------------------------|
| F-niel-001 | Process epxlorer.exe (PID 9920) was spawned by explorer.exe (PID 5116) at 2025-07-22 18:53:11 UTC, just 2 minutes 21 seconds after user login. The binary was executed from C:\Users\award\Downloads\epxlorer.exe.           | CRITICAL<br>HIGH conf    | T1036.005<br>T1204.002 | volatil:<br>2.7.0 +<br>MemProch<br>5.17.8 |
| F-niel-002 | epxlorer.exe established an outbound HTTPS connection to 104.21.1.247:443 at 2025-07-22 19:12:33 UTC — 19 minutes after execution. The destination IP is a Cloudflare edge node fronting the attacker-controlled domain api- | CRITICAL<br>HIGH conf    | T1102.002<br>T1573     | volatil:<br>2.7.0                         |

| ID         | FINDING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SEVERITY /<br>CONFIDENCE                 | MITRE<br>ATT&CK                           | TOOL                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|-------------------------------------------|
|            | telegram-org.ctf.do. The malware uses Telegram Bot API (token: 8169267144:AAFwkHmXh71SMVcvFLantjTlwRbT9HdJdU) as its C2 channel, abusing legitimate cloud infrastructure to blend with normal HTTPS traffic.                                                                                                                                                                                                                                                                                          |                                          |                                           |                                           |
| F-niel-003 | <p>epxplorer.exe was discovered at three virtual addresses in the memory filescan, including a partial browser download artifact (epxplorer.exe435568.crdownload).</p> <p>The .crdownload extension is specific to Chromium-based browsers (Microsoft Edge in this environment — confirmed by extensive Edge artifacts throughout the filescan). Per CTF intelligence, the phishing email originated from ktso.sec@mailforce.net.</p>                                                                 | <div>HIGH</div> <div>HIGH conf</div>     | <div>T1566.001</div> <div>T1204.002</div> | volatil:<br>2.7.0                         |
| F-niel-004 | <p>The compromised host (TSO-ATC-CT412, user 'award') runs ATC_RADAR.exe — an air traffic control radar application with multiple active processes and network connections to 8.8.8.8 (Google DNS-over-HTTPS). The workstation name (ATC-CT412) and application context confirm this is not a standard enterprise endpoint — it is an aviation operations system. Multiple ATC_RADAR.exe instances ran concurrently with epxplorer.exe, meaning the malware coexisted with active ATC operations.</p> | <div>CRITICAL</div> <div>HIGH conf</div> | <div>T1003.001</div>                      | volatil:<br>2.7.0 +<br>MemProch<br>5.17.8 |



| TYPE    | INDICATOR                                                        | CONTEXT                  | F      |
|---------|------------------------------------------------------------------|--------------------------|--------|
| SHA-256 | 3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35 | Evidence<br>archive hash | E      |
| SHA-256 | 7b45e797acabaf592784bf7efceb979e3f036d5b9dd47f52280c17e38864c2dd | Memory<br>dump hash      | E      |
| MD5     | 75ea94b54420c39dcd3d8ce574ba9d34                                 |                          | F<br>n |

| TYPE         | INDICATOR                              | CONTEXT                                                  | F      |
|--------------|----------------------------------------|----------------------------------------------------------|--------|
|              |                                        | Malware hash<br>(epxplorer.exe)                          |        |
| Filename     | epxplorer.exe                          | Malware<br>binary —<br>masquerades<br>as<br>explorer.exe | F<br>n |
| File<br>Path | C:\Users\award\Downloads\epxplorer.exe | Malware<br>execution<br>path                             | F<br>n |
| Filename     | epxplorer.exe435568.crdownload         | Browser<br>partial<br>download<br>artifact<br>(Edge)     | F<br>n |
| IP           | 104.21.1.247                           | C2 server<br>(Cloudflare<br>proxy)                       | F<br>n |
| Domain       | api-telegram-org.ctf.do                | Attacker-<br>controlled<br>Telegram Bot<br>API C2        | F<br>n |
| Email        | ktso.sec@mailforce.net                 | Phishing<br>email sender                                 | F<br>n |
| IP           | 172.17.4.12                            | Compromised<br>host internal<br>IP                       | F<br>n |



EVID-001 — Be'lkaCTF\_7\_CASE250722\_KTS0AERO.zip

SHA-256: 3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35

Source: [https://dl.ctf.do/BelkaCTF\\_7\\_CASE250722\\_KTSAERO.zip](https://dl.ctf.do/BelkaCTF_7_CASE250722_KTSAERO.zip)

| TOOL           | VERSION  | RUNTIME   | VALIDATED   |
|----------------|----------|-----------|-------------|
| volatility3    | 2.7.0    | docker    | ✓           |
| memprocfs      | 5.17.8   | host_fuse | ✓           |
| plaso          | 20240512 | docker    | ✓           |
| mft-tools      | 1.2.0.0  | docker    | ✓           |
| sift-vm        | ?        | ?         | ✓           |
| session_canary | ?        | ?         | ⚠️ DEGRADED |



| PROPERTY     | VALUE                                                        |
|--------------|--------------------------------------------------------------|
| Hostname     | TS0-ATC-CT412                                                |
| Username     | award                                                        |
| Os           | Windows 10 build 19041<br>(19041.1.amd64fre.vb_release.1912) |
| Architecture | x64                                                          |
| Processors   | 4                                                            |
| Host Ip      | 172.17.4.12                                                  |
| Dump Time    | 2025-07-22 19:18:53 UTC                                      |
| Kernel Base  | 0xf80617e00000                                               |

|                 |                                     |
|-----------------|-------------------------------------|
| Nt Product Type | NtProductWinNt                      |
| Source Tool     | volatility3 2.7.0 windows.info.Info |

Source: volatility3 2.7.0 windows.info.Info

Immediate containment and eradication steps per finding. Prioritize CRITICAL severity items first.

**● F-niel-001: Malware Execution: explorer.exe Masquerading as explorer.exe**

Immediately isolate TSO-ATC-CT412 from all networks. Kill PID 9920. Quarantine C:\Users\award\Downloads\explorer.exe. Do NOT reboot before capturing full disk image — volatile evidence in memory will be lost.

**● F-niel-002: Command and Control via Fraudulent Telegram Bot API**

Block outbound HTTPS to api-telegram-org.ctf.do at network perimeter. Block Telegram Bot API endpoints if not business-justified. Deploy SSL inspection for traffic to unknown Cloudflare IPs. Check firewall logs for additional connections from 172.17.4.12 in the 18:50-19:18 UTC window.

**● F-niel-003: Phishing Delivery Vector: Browser Download of Malicious Executable**

Search mail server for emails from ktso.sec@mailforce.net — identify all recipients. Purge the phishing email from all inboxes. Block mailforce.net at email gateway. Initiate user awareness training for 'award' and ATC personnel. Review Edge browser security policies — consider disabling automatic execution of downloaded files.

**● F-niel-004: Critical Asset Compromised: Air Traffic Control Workstation**

Escalate to aviation safety officer and CISO immediately. This may require mandatory reporting to the national aviation authority. Ground the workstation — do not allow it to process live radar data until forensic imaging is complete and the system is rebuilt from trusted media. Engage the ATC software vendor (ATC\_RADAR.exe) to verify software integrity. Review separation requirements between administrative workstations and operational ATC systems.



The following areas were not examined and represent uncertainty in the assessment. Each gap should be addressed before finalizing conclusions.

**⚠ Disk Image Not Analyzed**

No disk image was available. Without filesystem forensics (Sleuth Kit, MFT analysis), we cannot confirm persistence mechanisms (registry Run keys, scheduled tasks, WMI subscriptions), determine if additional files were dropped, or establish a full file-system timeline.

**⚠ Network Capture Unavailable**

No PCAP was provided. Encrypted C2 traffic cannot be decrypted or analyzed for data exfiltration volume. Lateral movement within the 172.17.4.0/24 subnet cannot be assessed.

**⚠ Registry Hives Not Examined**

No registry analysis was performed (RegRipper available but no hives extracted). User activity artifacts (RecentDocs, UserAssist, Shellbags) and malware persistence locations (Run, RunOnce, Services) remain unexamined.

**⚠ Malware Not Reverse-Engineered**

explorer.exe was extracted from the memory dump but not statically or dynamically analyzed. Full capability assessment (keylogging, screen capture, lateral movement modules, data exfiltration mechanisms) is pending.

**⚠ Supplementary Evidence Not Analyzed**

The 5.6 GB supplementary file (BelkaCTF\_7\_CASE250722\_EXH250723-2.zip, SHA-256: e42f5254a058721f2cabeed2219f0e0783419addb094a0fd8f07f75818556fe7) was not downloaded or analyzed. This file contains Android artifacts and additional evidence per the CTF documentation.

**⚠ Single Host Analysis Only**

Only one memory dump was analyzed. The compromise of additional systems on the network cannot be ruled out without broader endpoint telemetry.



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This report is auto-generated from structured case data. All findings require human review before finalization.